Machine Learning-Based Flight Delay Prediction

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◆ Introduction

Flight delays are a common issue in air transportation, affecting millions of passengers and airline operations. Delays can lead to increased costs, disrupted schedules, and reduced customer satisfaction.

This project aims to analyze and predict flight arrival delays using multiple data sources, including flight records, airport information, and weather data such as temperature and precipitation. By integrating operational and environmental factors, we explore how these variables influence delay patterns.

Through data analysis and machine learning models, this study seeks to identify key factors driving delays and evaluate the performance of different predictive approaches.

◆ Data sources

Data Source	Description	Processing Approach	Purpose
Flight Data	Historical flight records including departure time, arrival delay, distance, and airline information	Filtered Jan–Mar 2014 data; removed missing values and extreme delays; selected key variables for modeling	Capture operational factors influencing flight delays
Airport Data	Airport metadata including IATA codes and geographic coordinates (latitude, longitude)	Merged with flight data by origin and destination; used to compute great-circle distance	Incorporate geographic features and calculate flight distance
Weather Data (NOAA)	Daily weather data including temperature (TMAX) and precipitation (PRCP)	Retrieved via API; transformed from long to wide format; merged with flights by date	Analyze environmental impact on flight delays

Final dataset contains approximately 35,000+ observations and 13 features after preprocessing and filtering.

◆ Feature Description

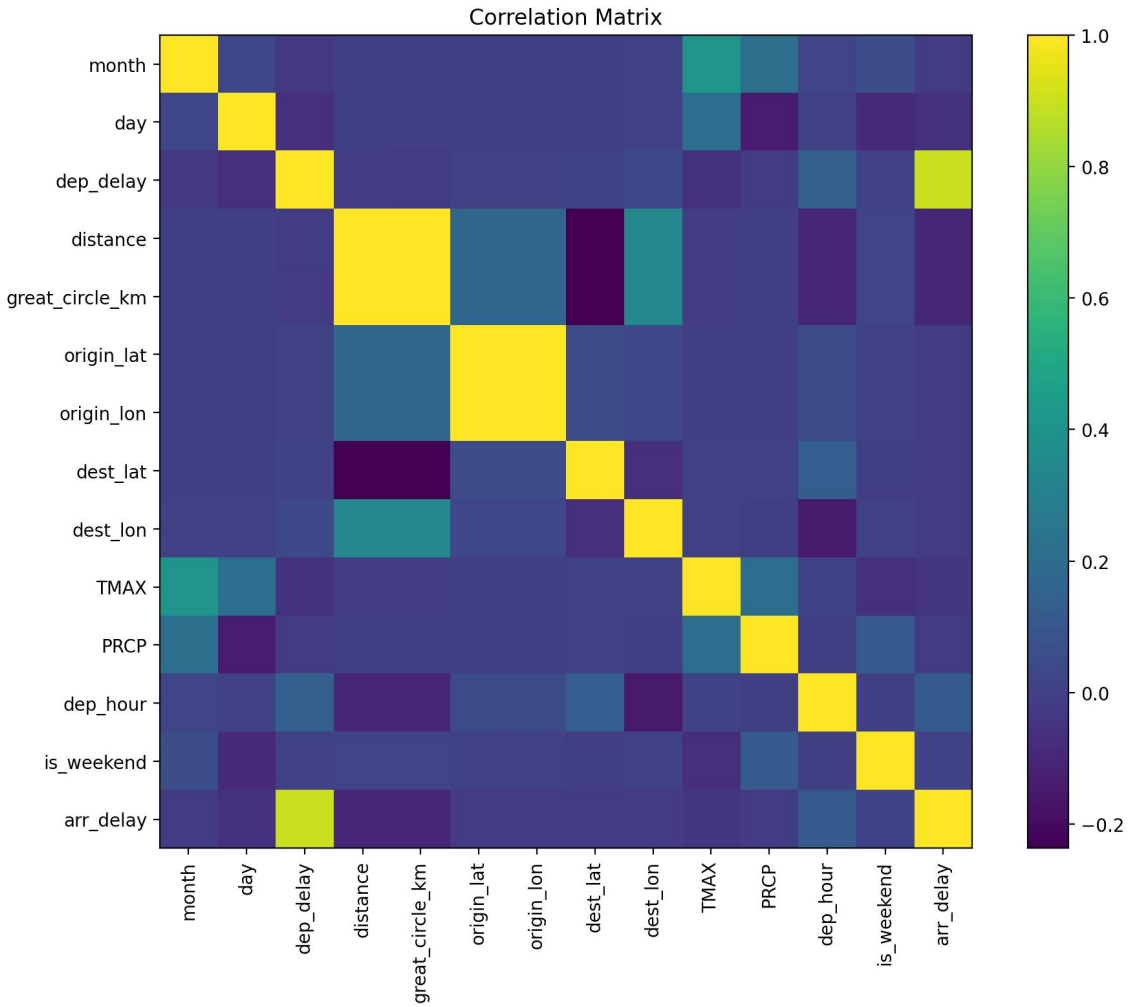
Feature	Description	Type
dep_delay	Departure delay in minutes	Numeric
distance	Flight distance between origin and destination (km)	Numeric
TMAX	Maximum daily temperature at departure airport	Numeric
PRCP	Daily precipitation at departure airport	Numeric
dep_hour	Hour of scheduled departure	Numeric
is_weekend	Whether the flight is on a weekend (0/1)	Categorical
origin_lat / lon	Geographic coordinates of origin airport	Numeric
dest_lat / lon	Geographic coordinates of destination airport	Numeric
Feature	Description	Type

◆ Summary of the results

Correlation Matrix

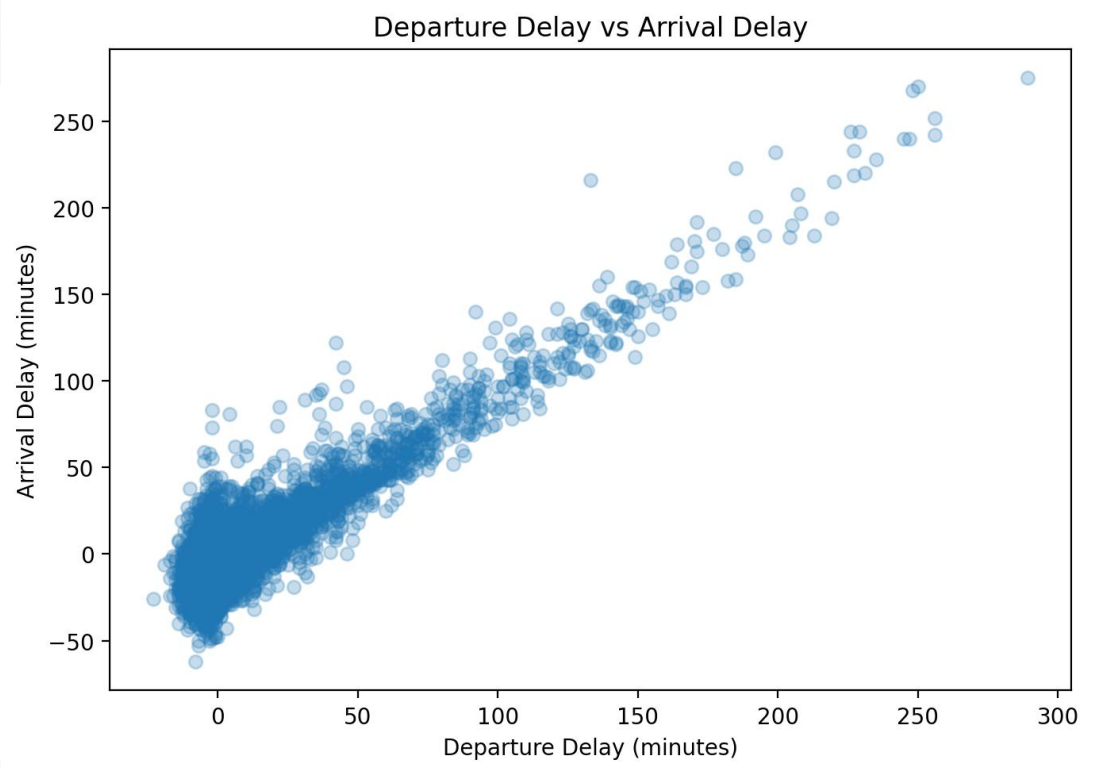
This heatmap shows the relationships between different features and arrival delay.

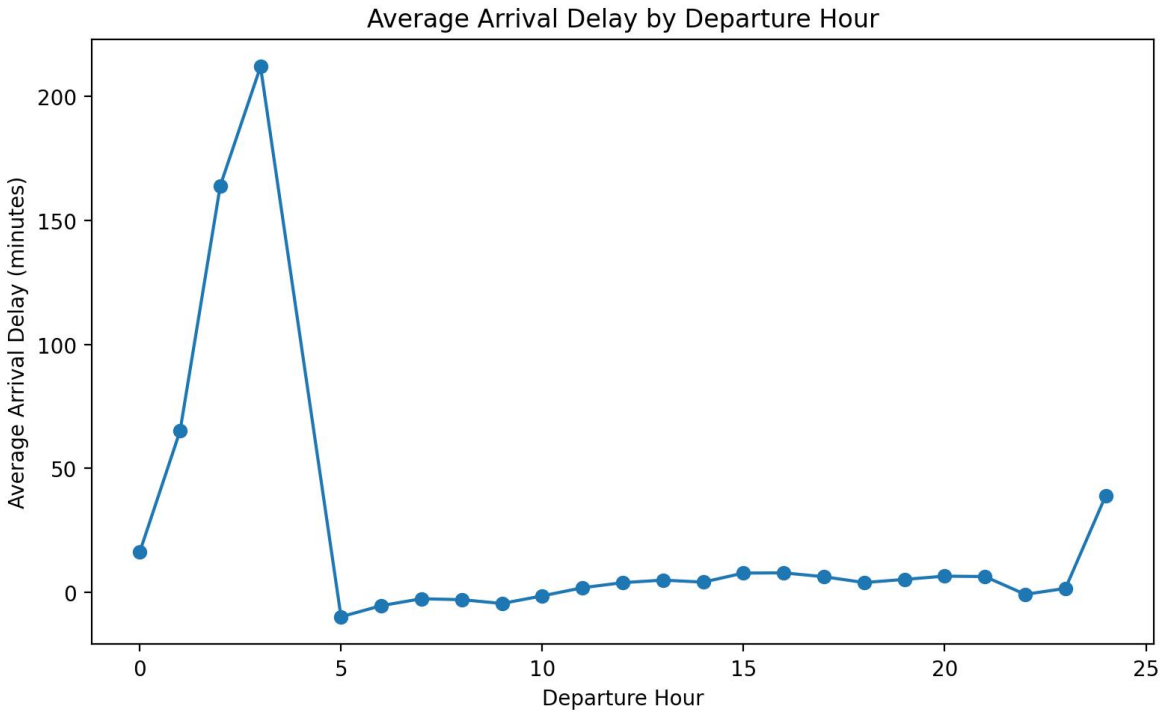
We observe that departure delay has the strongest correlation with arrival delay, while other features such as weather and distance show weaker but still useful relationships.



◆ Summary of the results

This scatter plot shows a strong positive correlation between departure delay and arrival delay.
Flights that depart late are very likely to arrive late as well.





This line chart shows how average arrival delay varies by departure hour.

We observe that delays tend to increase during certain time periods, indicating that time-related factors can influence flight delays.

◆ Summary of the results

Model	MAE	RMSE	R^2
Linear Regression	8.31	11.44	0.84
Random Forest	7.46	10.27	0.87
XGBoost	7.04	9.81	0.88

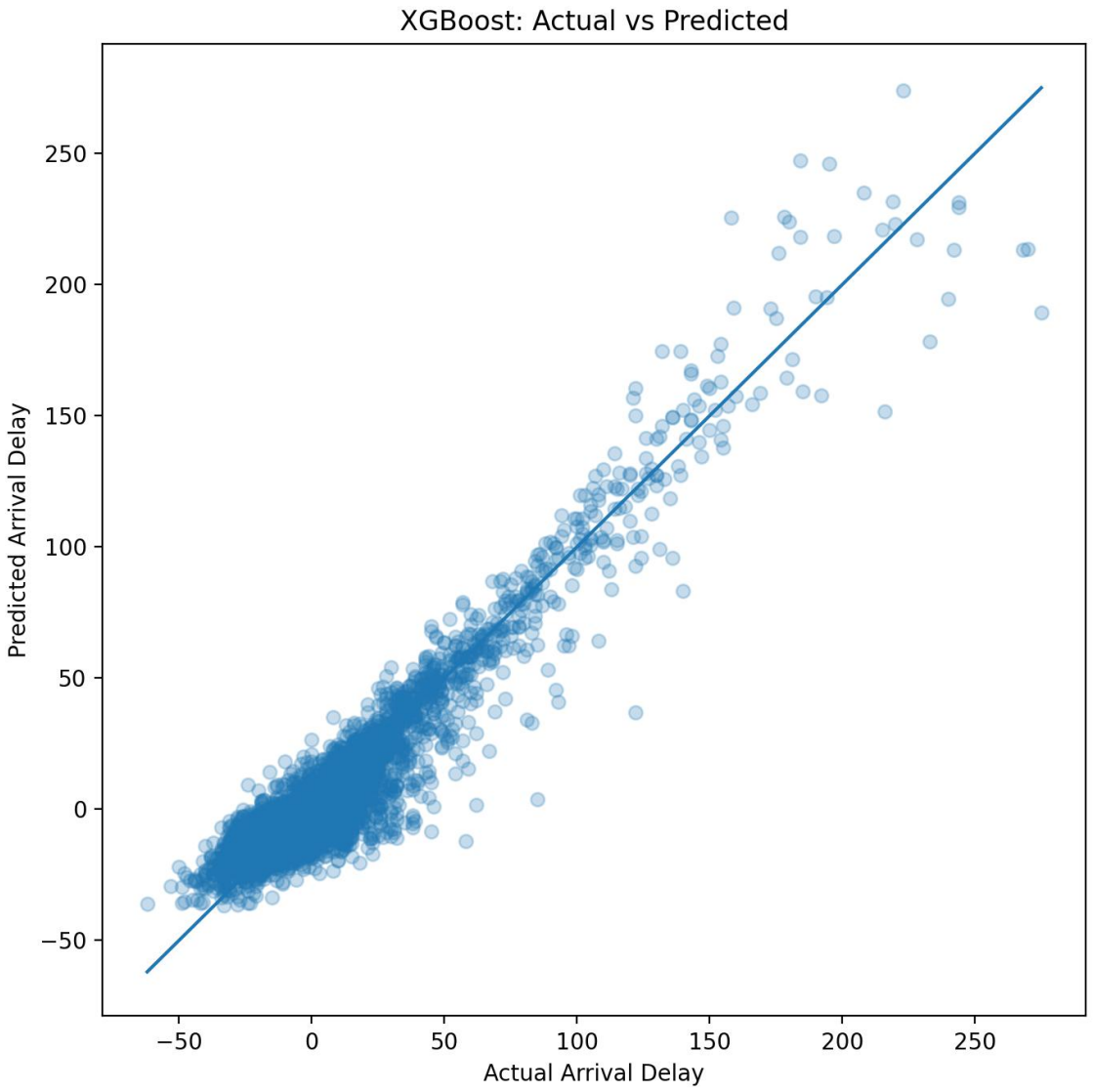
We compare three models: Linear Regression, Random Forest, and XGBoost.
Random Forest significantly improves over Linear Regression, reducing both MAE and RMSE.
XGBoost further improves performance and achieves the best results across all metrics, with the lowest errors and highest R^2.
The R^2 value of 0.88 indicates that the model explains most of the variance in arrival delays.
These improvements come from the ability of tree-based models to capture non-linear relationships in the data.
The improvement between Random Forest and XGBoost is smaller, suggesting diminishing returns as model complexity increases.

◆ Summary of the results

This plot compares actual and predicted arrival delays.
Most points lie close to the diagonal line, indicating good prediction accuracy.

Some larger deviations appear for extreme delays, suggesting that very large delays are harder to predict.

The model performs well overall, especially for typical delays, but struggles slightly with extreme cases.



◆ Challenges

Data quality issues such as missing values and inconsistencies required careful cleaning and preprocessing.

Integrating multiple data sources, including flight, airport, and weather data, was complex and required alignment by time and location.

Many features showed weak correlations with arrival delay, making it harder to improve model performance.

Extreme delay cases were difficult to predict accurately, as they are often influenced by unpredictable factors.

Thank you!